

Day 20 - Lecture 20.15: Mean, Median & Mode

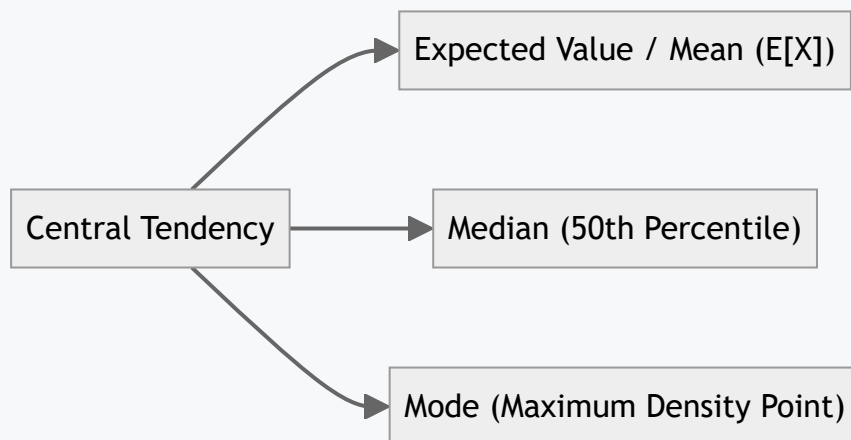
 Python 3.10+

 notes.pdf

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1. Overview & Measures of Central Tendency

Central tendency identifies the single central point or typical representative value within a probability distribution or dataset.



2. Mathematical Formalism

Expected Value (Mean μ):

The probability-weighted average of all possible values.

Discrete Random Variable:

$$\mathbb{E}[X] = \mu = \sum_{x \in \mathcal{X}} x \cdot p(x)$$

Continuous Random Variable:

$$\mathbb{E}[X] = \mu = \int_{-\infty}^{\infty} x \cdot f(x), dx$$

Linearity of Expectation (Crucial Machine Learning Property):

For any random variables X and Y and constants $a, b, c \in \mathbb{R}$, regardless of statistical independence:

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

Median (m):

The value that splits the probability mass into two equal halves:

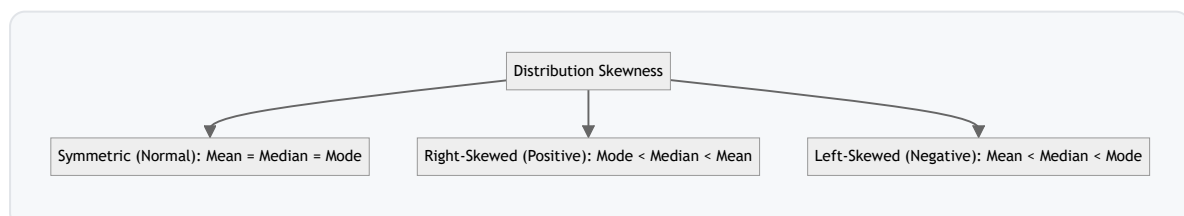
$$P(X \leq m) \geq 0.5 \quad \text{and} \quad P(X \geq m) \geq 0.5$$

Mode:

The value at which the probability function attains its absolute peak:

$$\text{Mode}(X) = \arg \max_x p(x) \quad \text{or} \quad \arg \max_x f(x)$$

3. Impact of Distribution Skewness



In right-skewed data (e.g., household incomes, transaction values, web latency), extreme outlier values pull the Mean upward, making the **Median** a more robust metric for central tendency.

4. Python Implementation

```
import numpy as np
from scipy import stats

# Generate right-skewed lognormal data (e.g., web server response times)
np.random.seed(42)
latencies = np.random.lognormal(mean=2.0, sigma=0.8, size=100_000)

mean_val = np.mean(latencies)
median_val = np.median(latencies)
mode_val = stats.mode(np.round(latencies, 1), keepdims=True).mode[0]

print(f"Server Latency Metrics (Right-Skewed):")
print(f"  Mode:    {mode_val:.2f} ms")
print(f"  Median:  {median_val:.2f} ms")
print(f"  Mean:    {mean_val:.2f} ms")
print(f"Ordering Holds (Mode < Median < Mean): {mode_val < median_val < mean_val}")
```

5. Key Takeaways & Interview Points

- **Linearity of Expectation Holds Universally:** $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ requires **zero** assumptions about independence.
- **Robustness in Production:** Mean is sensitive to outliers; Median is immune to extreme values below the 50th percentile.